Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/08/2005
PI Name:	Abbud-Madrid, Angel Ph.D.		
Project Title:	FUNDAMENTAL EVALUATION OF WATER MIST FOR FIRE SAFETY IN SPACECRAFT AND EXTRATERRESTRIAL ENVIRONMENTS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Delplanque, Jean (Colorado School of Mines) McKinnon, Julian (Colorado School of Mines)		
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Task Description:	This investigation is intended to provide an integrated suite of tasks that involve both experimental studies and detailed modeling of fine water mist in fire suppression scenarios relevant to spaceflight needs. All research on water mist will be done in comparison to a baseline of CO2, the suppression agent of choice onboard the ISS and the most abundant component in the Martian atmosphere. The research plan is based on a hierarchical model. First, a study of simplified scenarios is performed at a detailed level. Second, the critical elements of the detailed model are extracted and then encapsulated in a reduced-order manner for inclusion in the next higher-level scenario. The process is continued until a robust understanding of fire suppression at the level of an ISS rack is obtained. The experiments will be performed in both normal and reduced gravity for an accurate validation of the numerical model and to provide observations of fire suppression behavior under extraterrestrial conditions. The overall goal of this fundamental study and the resulting numerical code, validated with experiments, is to evaluate the role of water mist in the next generation of fire safety systems on spacecraft and habitation modules on the surface of other planets. This project will simultaneously satisfy two needs – scientific and NASA programmatic. At the scientific level, the combined experiment/modeling program will lead to a basic understanding of water mist and CO2 fire suppression in realistic fire scenarios (e.g., wire bundles, burning cloth, etc). A determination will be made of the interaction of reduced oxygen concentration and lowered temperature with smoldering rates and smoke generation. This basic understanding will advance the field of fire suppression for both spacecraft and terrestrial applications. At the NASA programmatic level, a series of sub-models (fire source, suppression agent generation and distribution, and radiation models) will be developed resulting on an integrated model that can be employed by NASA as design aids for spaceflight hardware. Deliverables will also fall into scientific and programmatic categories. In the scientific category there will be studies in the form of peer-reviewed journal articles on the basic understanding gained in the sub-models and the issues resolved in model integration. In the programmatic category there will be software modules at the sub-model and integrated model level, and data that will be available to investigators of reduced gravity flame spread, and fire growth and suppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results on the effectiveness of the various suppressant agents used in this project will serve to determine an optimum fire-fighting technique for space and for Earth applications. While the main emphasis is on fighting fires on spacecraft, the results will be generalized for all gravity conditions, including Earth's gravity. The software code developed will serve as a tool for designing fire suppression systems for aircraft, ships, museums, libraries, and a variety of enclosed spaces.
Task Progress:	During FY2005, the investigation focused on the study of water mist as a suppressant agent in electrical fires under normal-gravity conditions. The experimental work concentrated on evaluating the suppression effects of various droplet size distributions and the behavior of water mist in a constrained geometry. In a parallel effort, the numerical work focused on the simulation of the generation, evaporation, and distribution of the mist as it moves through the container and as it interacts with the fire source. Direct and indirect water-mist injection with a high-momentum jet are used with various droplet-size distributions and flow rates to extinguish a burning wire inside a container similar to the Space Shuttle mid-deck locker. The wire is located 40 cm from the water mist injector and for the indirect-injection tests a 13-cm wide, 25-cm high baffle is placed at 26 cm from the nozzle, in front of the burning wire. This last configuration is used to provide an extremely difficult path for the water mist to reach the burning wire. Interestingly, varying the width of the baffle from 5 to 13 cm had only a minor effect on the suppression efficacy of the high-momentum jet. Different droplet-size distributions and flow rates are possible by varying the water pressure on a water-mist injector with a 0.2-mm diameter orifice. Droplet size distributions with a Sauter mean diameter (SMD) of 40 to 27 microns are achieved with pressures varying from 100 to 1000 psi, respectively. Water mist quickly extinguishes a fire that is directly impacted by the droplets. However, much longer spraying times and larger amounts of water are required to suppress fires burning behind a baffle. Extinction times and water amounts are reduced as the droplet size distribution is lowered. A numerical model simulating the generation, evaporation, and distribution of a polydispersed spray accurately captures the phenomena observed experimentally. The predicted behavior of the water mist blocked by the baffle shows no direct interaction between the mist and the burning wire. The droplets do not have significant inertia to move around the obstacle and instead a large recirculation region is generated, where the mist will eventually impinge on the chamber walls. In the case where the water mist is accompanied by a 5 m/s coflow, the droplets get diverted and move around the obstacle, but even under these conditions, the reattachment length for the spray occurs past the chamber wall, making it impossible to have direct interaction with the wire. The experimental results show that the wire will interact with only about 1.5% of the mist by mass. The discretization process associated with the numerical model is not accurate to that low percentage level. Increasing the number of representative droplets would increase the model accuracy and might allow the description of this interaction. In the current scenarios, the lack of interaction would suggest, as confirmed by experiments, that fire suppression effects would be minimal, and they would mainly be due to mechanisms other than enthalpy extraction. This experimental-numerical comparison has provided valuable information in the on-going validation of the mist dispersion sub-model, which is part of a three sub-model suite (along with the fire and radiation sub-models) required for the final development of a reduced order model to describe the complete fire suppression process. As a result of the ineffectiveness of high-momentum, large-droplet-size spray jets to maneuver around obstacles and extinguish obstructed fires, the research effort is now shifting to the study of low-momentum, ultra-fine mist as a technique to produce total flooding of the confined space and effective suppression even in complicated geometries with fires hidden by multiple obstacles. For these tests, the high-pressure nozzles are replaced by an ultrasonic mist generator, which generates a droplet size distribution with a SMD lower than 10 microns at flow speeds in the vicinity of 0.5 m/s. From the few cases run to date, it is clear to see that ultra-fine mist can easily go around the baffle and rapidly flood the entire container in a gas-like manner. For the same experimental conditions described for the indirect-injection case with a baffle, the flame propagating down the insulated wire was extinguished by the ultra-fine mist with an average of 4 ml of water in approximately 10 seconds, an order of magnitude lower, in both extinction time and water amount, than the high-momentum, large-droplet-size spray jet. These promising results have prompted an evaluation of ultra-fine water mist as a potential agent for the suppression of the fire types most likely to be encountered in spacecraft applications. An experimental and numerical feasibility study of the fire suppression properties of ultra-fine water mist will be an important part of the next phase of the project.

Bibliography Type:	Description: (Last Updated: 03/21/2008)
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Presentation	Abbud-Madrid, A.; McKinnon, J. T.; and Delplanque, J. P. "Fundamental Evaluation of Water Mist for Fire Safety in Spacecraft" Microgravity Fire Modeling Working Group Meeting Apr-2005